

Exp no :08

Date :

Using Process Explorer to Identify Suspicious Processes

Description

- Process Explorer is a part of the **Microsoft Sysinternals Suite**. It is a powerful tool used to view detailed information about system processes. It helps investigators and administrators analyze active processes, detect suspicious behavior, monitor CPU and memory usage, and verify process authenticity using digital signatures.

Step-by-Step Process

Step 1: Download and Setup Process Explorer

1. Go to the official Microsoft Sysinternals website:
 <https://learn.microsoft.com/en-us/sysinternals/downloads/process-explorer>
2. Click **Download Process Explorer**.
3. Extract the downloaded ZIP file to a folder.
4. Right-click procexp64.exe (for 64-bit) or procexp.exe (for 32-bit) → select **Run as Administrator**.

Step 2: Understand the Interface

1. The main window displays all running processes in a **hierarchical tree view**.
2. Each process shows details such as **PID, CPU usage, memory usage, and company name**.
3. Color codes represent process states:
 -  **Green** — Newly started processes
 -  **Red** — Terminated processes
 -  **Light Blue** — Processes running under the current user
 -  **Pink** — Suspended processes

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Secure System		172 K	64,896 K	188		
Registry	0.12	8,184 K	54,140 K	228		
System Idle Process	74.01	60 K	8 K	0		
System	1.67	68 K	11,360 K	4		
Interrupts	1.31	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		1,160 K	1,420 K	888		
Memory Compression	< 0.01	2,404 K	10,02,652 K	3956		
csrss.exe		2,572 K	6,728 K	1276		
wininit.exe		1,732 K	8,872 K	1364		
services.exe	0.83	7,808 K	20,904 K	1512		
svchost.exe	< 0.01	15,000 K	44,172 K	1672	Host Process for Windows S...	Microsoft Corporation
WmiPrvSE.exe		4,656 K	14,888 K	7204	WMI Provider Host	Microsoft Corporation
WmiPrvSE.exe	7.73	23,556 K	31,628 K	7212	WMI Provider Host	Microsoft Corporation
unsecapp.exe		2,328 K	11,108 K	7752	Sink to receive asynchronous...	Microsoft Corporation
StartMenuExperienceHos...	< 0.01	1,44,188 K	1,66,744 K	11404	Windows Start Experience H...	Microsoft Corporation
SearchHost.exe		86,132 K	1,36,876 K	11412		Microsoft Corporation
msedgewebview2.exe	< 0.01	43,944 K	1,16,864 K	13064	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		3,000 K	12,640 K	13168	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		15,556 K	41,260 K	13456	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		1,21,776 K	81,208 K	13472	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		9,804 K	18,396 K	13660	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		1,37,472 K	1,42,564 K	13836	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		8,868 K	20,556 K	17636	Microsoft Edge WebView2	Microsoft Corporation
WidgetBoard.exe		1,15,112 K	1,14,396 K	11668		Microsoft Corporation
RuntimeBroker.exe		19,436 K	77,324 K	11824	Runtime Broker	Microsoft Corporation
WidgetService.exe		5,924 K	29,100 K	6976	WidgetService.exe	Microsoft Corporation
unsecapp.exe		1,668 K	9,980 K	11864	Sink to receive asynchronous...	Microsoft Corporation
MicrosoftStartFeedProvid...		10,500 K	42,812 K	15304		Microsoft Corporation
RuntimeBroker.exe		2,436 K	14,168 K	15792	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		3,248 K	22,392 K	15784	Runtime Broker	Microsoft Corporation
TextInputHost.exe	0.36	2,32,896 K	2,72,072 K	15700		Microsoft Corporation
RuntimeBroker.exe		2,356 K	13,284 K	12512	Runtime Broker	Microsoft Corporation
IGCC.exe		44,144 K	59,684 K	8396	IGCC	Intel Corporation
secd.exe		4,560 K	19,504 K	18924	Apple Security Manager	Apple, Inc.
APSDaemon.exe		6,116 K	25,056 K	17304	Apple Push	Apple Inc.
ApplePhotoStreams.exe		8,932 K	41,740 K	21376	iCloud Photo Stream	Apple Inc.
ShellExperienceHost.exe	Susp...	62,008 K	87,020 K	22396	Windows Shell Experience H...	Microsoft Corporation
RuntimeBroker.exe		8,116 K	42,460 K	20896	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		2,456 K	14,232 K	2196	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		6,532 K	30,472 K	21816	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		8,036 K	26,006 K	21668	Runtime Broker	Microsoft Corporation

CPU Usage: 26.41% Commit Charge: 58.66% Processes: 319 Physical Usage: 65.24%

Step 3: Identify Suspicious Processes

1. Look for unfamiliar or oddly named processes (e.g., xkdjeo.exe, randomname123.exe).
2. Check the Company Name and Description:
 - Legitimate software usually shows known publishers like *Microsoft*, *Intel*, or *Adobe*.
3. Right-click the process → Properties → go to the Image tab.
4. Verify the Path of the executable file:
 -  Safe: C:\Windows\System32\
 -  Suspicious: C:\Users\<User>\AppData\Temp\ or Downloads\
5. Check for Digital Signature:
 - Valid signature = trusted developer
 - No signature or invalid = possibly malicious

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Step 4: Analyze Process Behavior

- Observe CPU, Memory, and I/O usage columns.
- If a small or unknown process consumes excessive CPU or memory, it may be malicious.
- Right-click the process → Properties → go to the TCP/IP tab.
- Check if it communicates with unknown external IP addresses.
- Examine Handles and DLLs tabs for suspicious loaded files or libraries.

Process	Image	Performance	Performance Graph	Disk and Network	GPU Graph	Threads	TCP/IP	Security	Environment	Job	Strings
explorer.exe	explorer.exe	144									
Process	Image	Performance	Performance Graph	Disk and Network	GPU Graph	Threads	TCP/IP	Security	Environment	Job	Strings
144	explorer.exe	144									
TID	CPU	Cycles Delta	Suspend Count	Start Address							
10724	< 0.01	28,63,233		shcore.dll!SHCoreThrea...							
10660				windows.immerseshell.s...							
12148				ntdll.dll!TpCallbackMayRu...							
12372				Windows.UI.Xaml.dll!C...							
8604				shcore.dll!SHCoreThrea...							
11768				ntdll.dll!TpCallbackMayRu...							
14556				ntdll.dll!TpCallbackMayRu...							
1804				shcore.dll!SHSelfValueW+							
9012				CoreMessaging.dll!CoreUI...							
12988				wlanapi.dll!WlanPrivateCa...							
13104				wlanapi.dll!WlanPrivateCa...							
2472				shcore.dll!SHCoreThrea...							
13952				wlanapi.dll!WlanPrivateCa...							
21040				ntdll.dll!TpCallbackMayRu...							
10880				ucrtbase.dll!ucrtbase!dr120							
3204				shcore.dll!SHSelfValueW+							
16816				shcore.dll!SHSelfValueW+							
7044				shcore.dll!SHCoreThrea...							
12324				shcore.dll!SHCoreThrea...							
12708				shcore.dll!SHCoreThrea...							
12308				ntdll.dll!TpCallbackMayRu...							
12312				ntdll.dll!TpCallbackMayRu...							
12300				ntdll.dll!TpCallbackMayRu...							

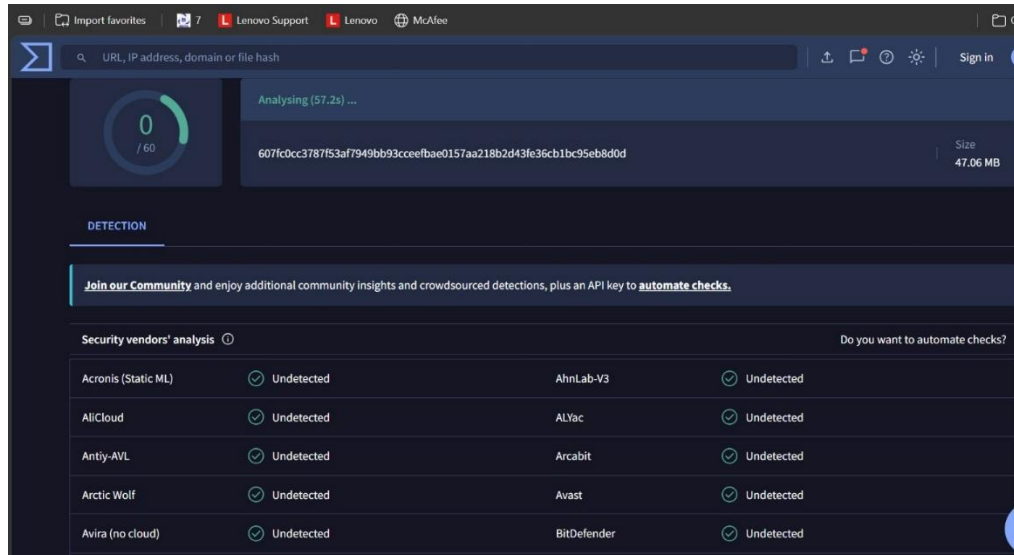
9976
19:49:20 21-09-2026
Wait:WlanUserRequest 8
0:00:07.812 12
0:00:12.812 Normal
95,533 5
52,60,80,17,360 2

Suspend
OK Cancel

Step 5: Verify Process Legitimacy

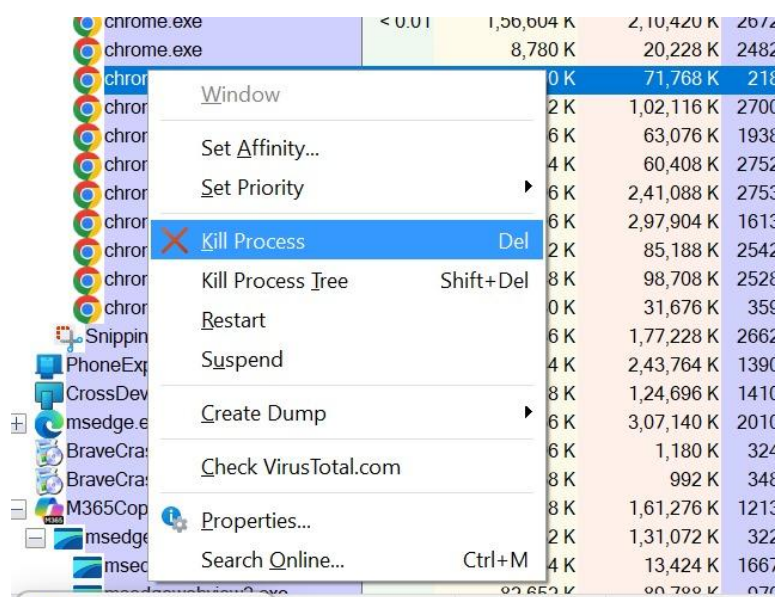
1. Search the process name on Google.
Example: svchost.exe vs svhst.exe (one letter missing — suspicious).

2. Visit <https://www.virustotal.com>
 - Upload the process file or search its name to verify if it's reported as malware.
3. Cross-check with **ProcessLibrary.com** or official vendor websites for authenticity.



Step 6: Take Appropriate Action

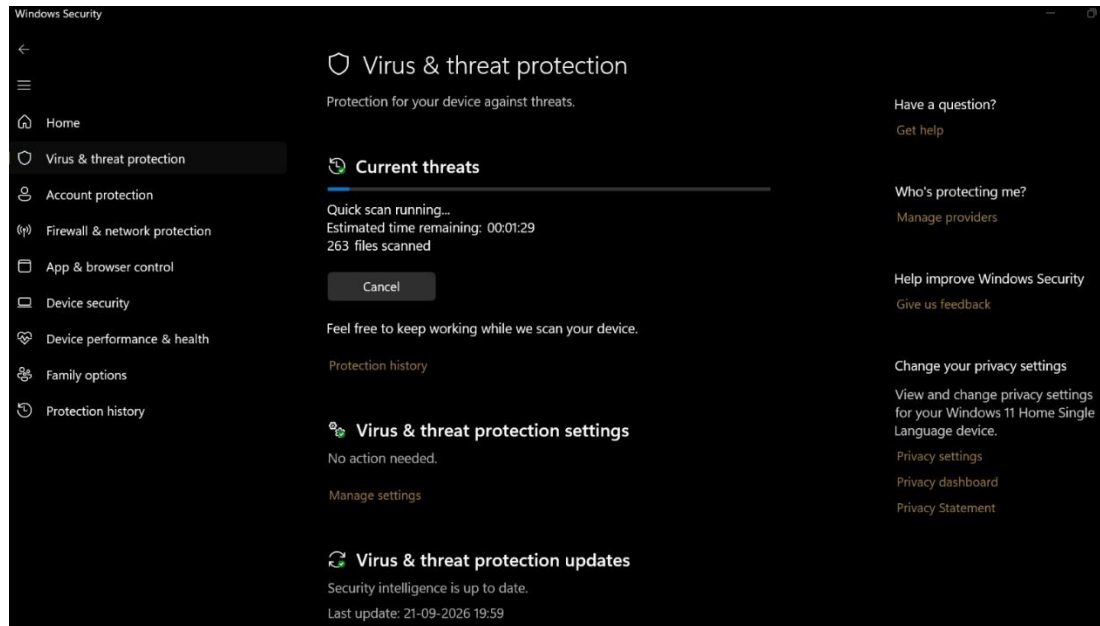
1. **If the process is confirmed malicious:**
 - Right-click the process → **Kill Process** to stop it.
 - Delete the corresponding executable file from its path.
2. **If unsure:**
 - Right-click → **Suspend Process** to stop it temporarily for investigation.
3. **After removal:**
 - Run a **Full System Scan** using Windows Defender or Malwarebytes to ensure no remnants remain.



Step 7: Example Observation

You find faangpath_simple_template.pdf consuming 70% CPU.

- **Path:** C:\Users\Admin\AppData\Temp\faangpath_simple_template.pdf
- **Digital Signature:** None
- **Company Name:** Unknown
- **Network Activity:** Shows connections to unknown IPs in the TCP/IP tab
- **Online Check:** VirusTotal confirms it as a **known trojan**
- **Action Taken:** Suspended → Killed → Deleted file → Performed full antivirus scan



Rubrics

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result

Using Process Explorer, suspicious processes were successfully identified by examining their **CPU usage, path, digital signature, and network activity.**

Confirmed malicious processes were terminated and removed to maintain system integrity.